Aim

To configure **IPsec (Internet Protocol Security)** on a network device/system to provide secure communication and protect data from unauthorized access, interception, and attacks.

Objectives

- To understand the concept of **IP Security (IPsec)**.
- To understand the need for securing IP-based communication.
- To study the components of IPsec.
- To understand the concept of a **VPN (Virtual Private Network)**.
- To understand **L2TP/IPsec** VPN configuration.
- To configure a VPN connection using **IPsec**.
- To understand the role of **CHAP and MS-CHAP v2** in authentication.
- To understand the use of a **Pre-Shared Key (PSK)** for IPsec authentication.
- To establish a secure VPN connection.

Theory

1. IP Security (IPsec)

- **IPsec (Internet Protocol Security)** is a collection of protocols used to secure communication over an IP network.
- It provides security at the **Network Layer (Layer 3)** of the OSI model.
- IPsec protects IP packets by providing **authentication, integrity, confidentiality, and secure key management**.
- It is commonly used to create secure **VPN connections** over public or untrusted networks such as the Internet.

Main Security Services of IPsec

Confidentiality

- Encrypts data so that unauthorized users cannot read it.

Integrity

- Ensures that the data has not been modified during transmission.

Authentication

- Verifies the identity of the communicating parties.

Protection Against Replay Attacks

- Helps prevent attackers from capturing valid packets and retransmitting them later.

2. Components of IPsec

IPsec mainly uses the following protocols and technologies:

Authentication Header (AH)

- **AH** provides authentication and integrity for IP packets.
- It helps verify that the packet came from the correct sender and was not modified.
- AH does **not provide encryption** of the actual data.

Encapsulating Security Payload (ESP)

- **ESP** provides confidentiality by encrypting the data.
- It can also provide authentication and integrity.
- ESP is widely used in modern IPsec VPN implementations.

Internet Key Exchange (IKE)

- **IKE (Internet Key Exchange)** is used to securely establish security associations and negotiate cryptographic parameters.
- It helps the communicating parties agree on encryption, authentication, and other security settings.

3. VPN (Virtual Private Network)

- A **VPN** creates a secure connection between two systems or networks over a public network such as the Internet.
- The data transmitted through the VPN is protected from unauthorized users.
- VPNs are commonly used for remote access and secure communication between different offices or networks.

4. L2TP/IPsec

- **L2TP (Layer 2 Tunneling Protocol)** is a tunneling protocol used to transport network traffic through a tunnel.
- L2TP itself does **not provide strong encryption**.
- Therefore, L2TP is commonly combined with **IPsec** to provide encryption and security.

L2TP/IPsec provides:

- **L2TP** → Creates the tunnel.
- **IPsec** → Provides security and encryption for the tunnel.
- **Authentication** → Verifies the user/device.
- **Pre-Shared Key (PSK)** → Can be used to authenticate the IPsec connection.

Thus:

L2TP + IPsec = Secure VPN Connection

5. CHAP

- **CHAP (Challenge Handshake Authentication Protocol)** is an authentication protocol used to verify the identity of a user or device.
- Instead of directly sending the password over the network, CHAP uses a **challenge-response mechanism**.

Working of CHAP

1. The server sends a challenge to the client.
2. The client calculates a response using the challenge and authentication information.
3. The client sends the response to the server.
4. The server verifies the response.
5. If the response is correct, authentication is successful.

6. MS-CHAP v2

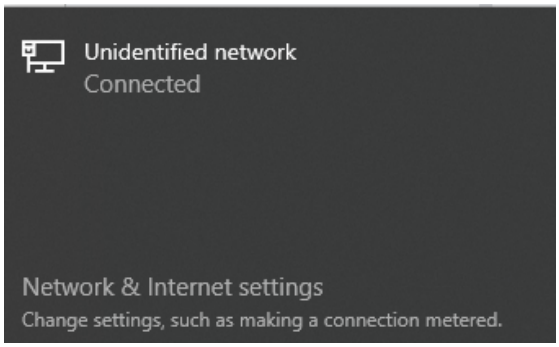
- **MS-CHAP v2 (Microsoft Challenge Handshake Authentication Protocol Version 2)** is an enhanced version of CHAP commonly used for Microsoft-based VPN authentication.
- It provides mutual authentication and is commonly supported by Windows VPN connections.

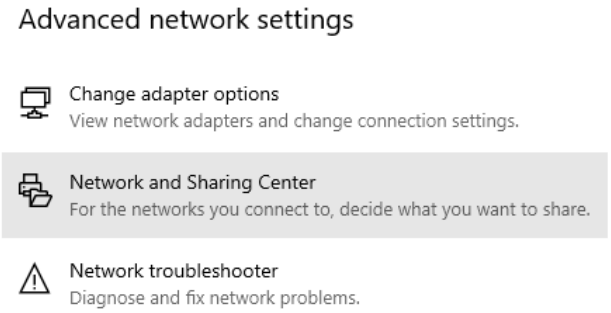
7. Pre-Shared Key (PSK)

- A **Pre-Shared Key (PSK)** is a secret value shared between the VPN client and VPN server before establishing the IPsec connection.
- The PSK is used to authenticate the communicating parties during the IPsec negotiation.
- The PSK should be kept secret and should not be shared with unauthorized users.

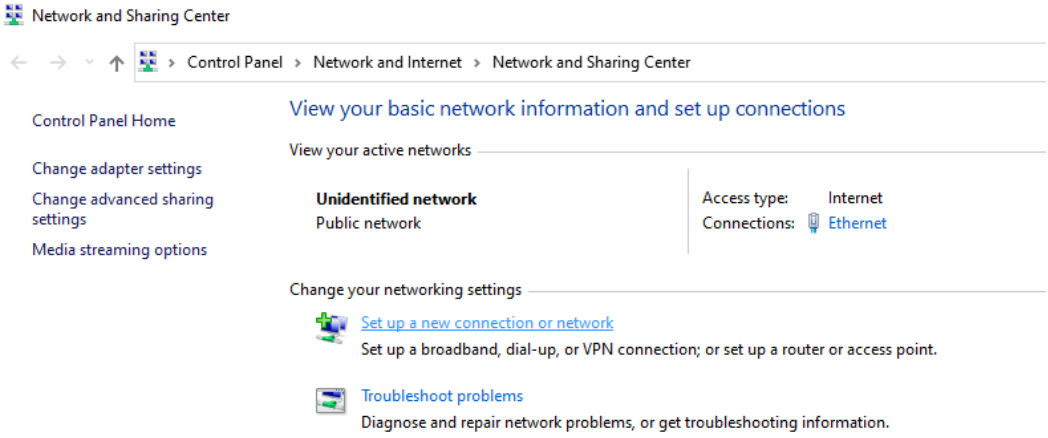
Practical Procedure

1. Select Open Network and Sharing Center. Or, if using Windows 10 version 1709 or newer, select Open Network & Internet settings, then click Network and Sharing Center.

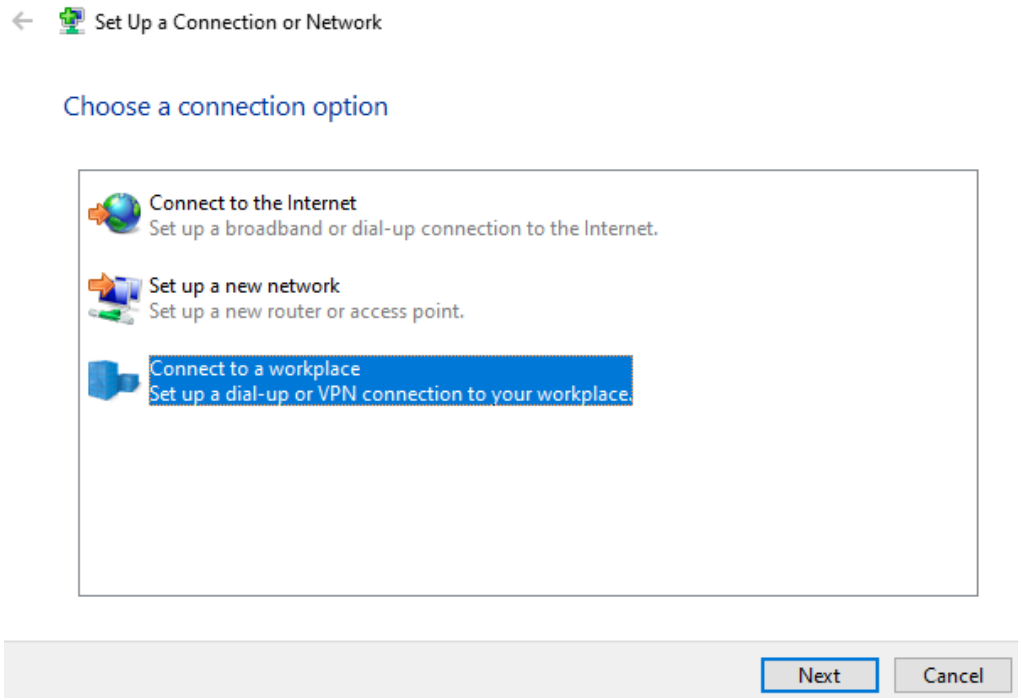




2. Click Set up a new connection or network.



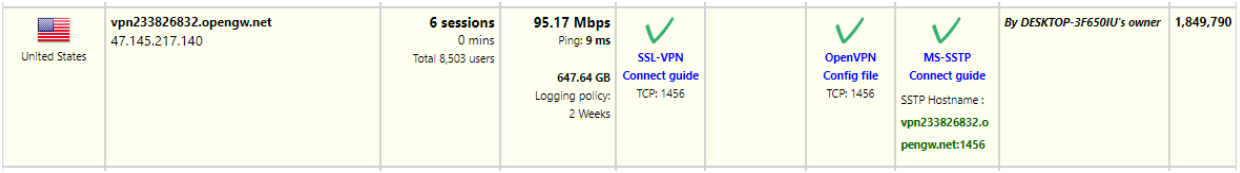
3. Select Connect to a workplace and click Next.



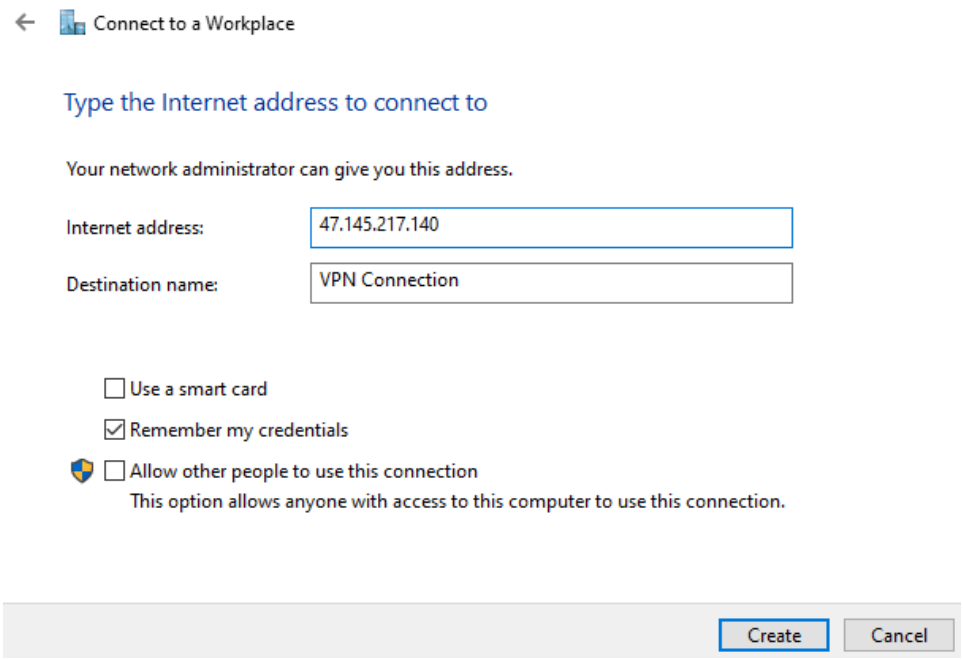
4. Click Use my Internet connection (VPN).



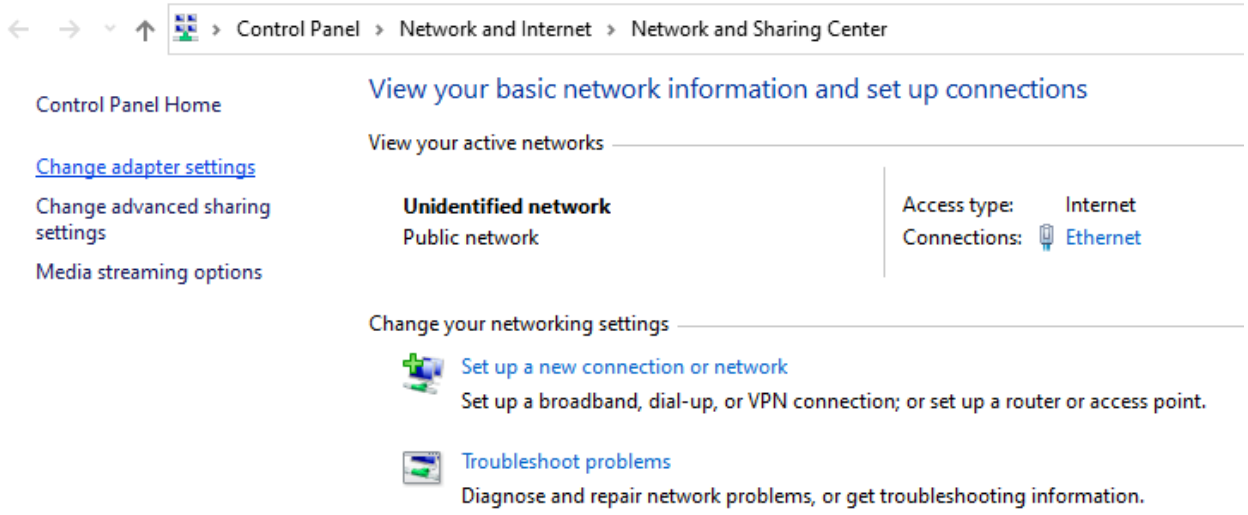
5. Enter Your VPN Server IP in the Internet address field. (www.vpngate.net)



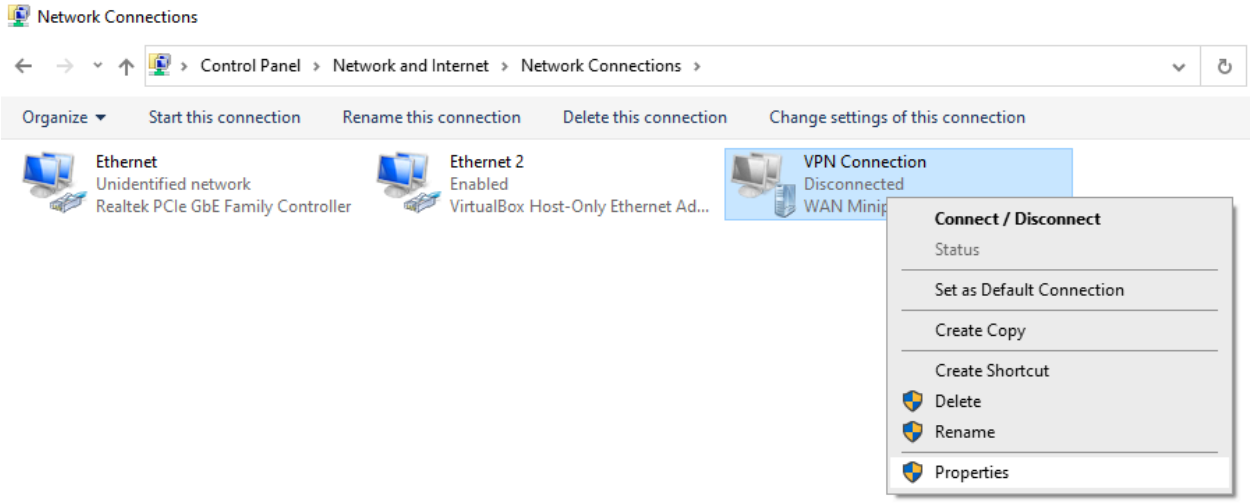
6. Enter anything you like in the Destination name field, and then click Create.



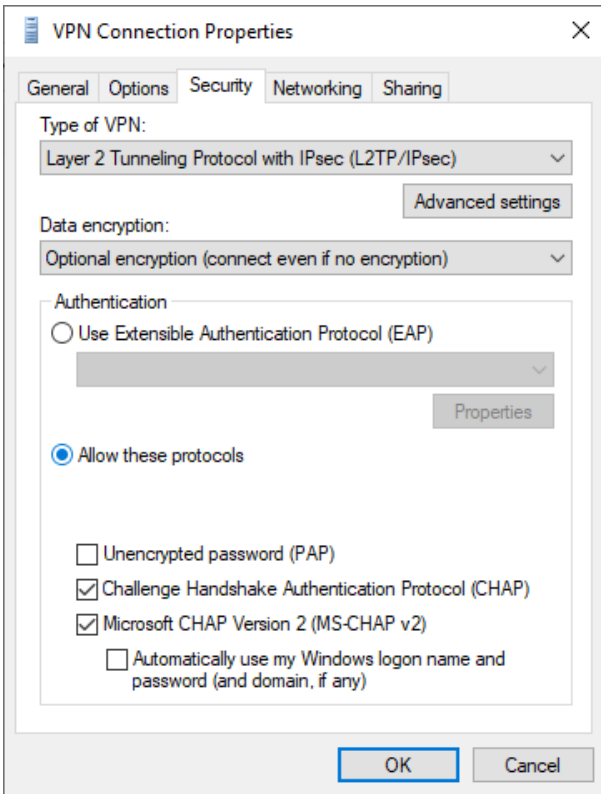
7. Return to Network and Sharing Center. On the left, click Change adapter settings.



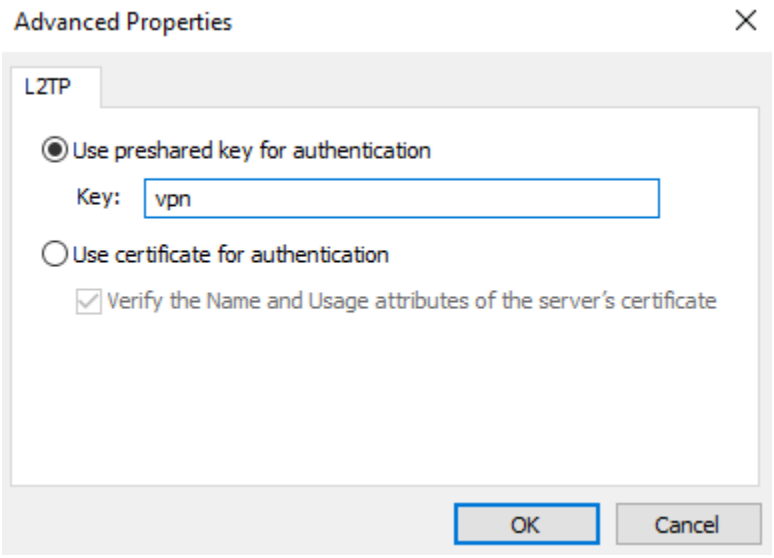
8. Right-click on the new VPN entry and choose Properties.



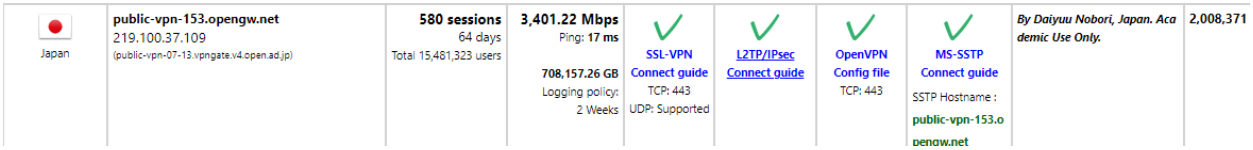
9. Click the Security tab. Select Layer 2 Tunneling Protocol with IPsec (L2TP/IPSec) for the type of VPN.
10. Click Allow these protocols. Check the Challenge Handshake Authentication Protocol (CHAP) and Microsoft CHAP Version 2 (MS-CHAP v2) checkboxes.



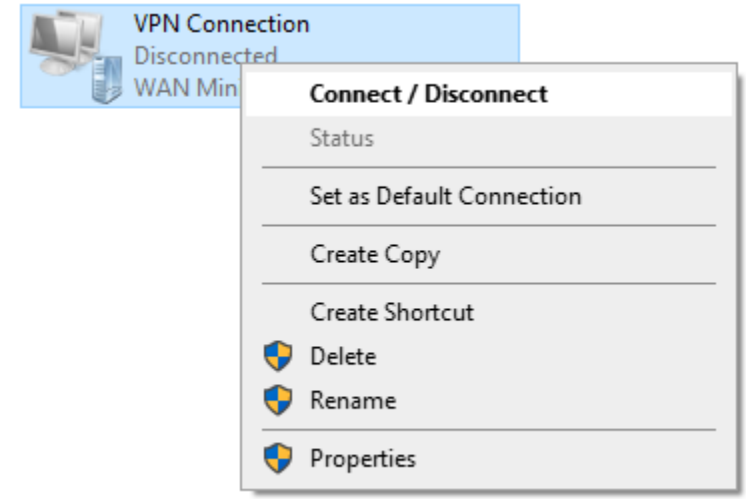
11. Click Advanced settings.

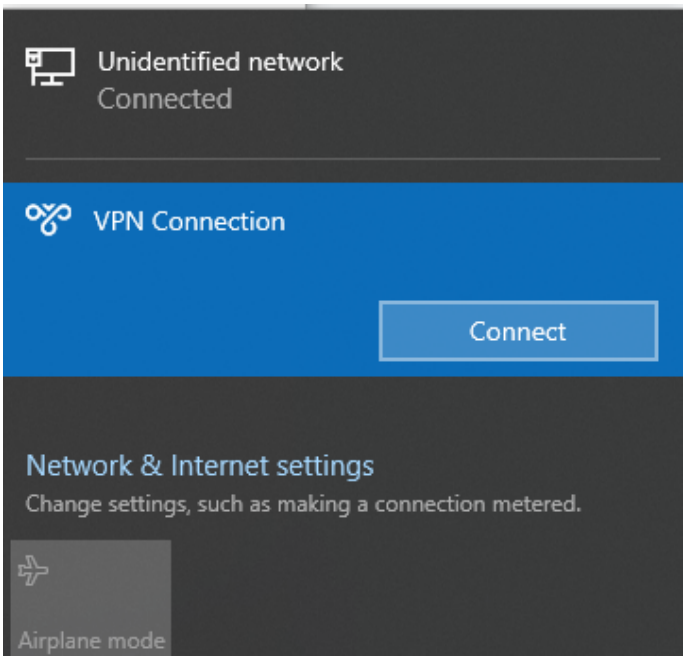


12. Select Use preshared key for authentication and enter Your VPN IPsec PSK for the key.



13. Click OK to close the Advanced settings.
14. Click OK to save the VPN connection details.





15. **Got Connected**

